Projectile Motion with Air Resistance

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1 Physical Model

1.1 Motivation

When we consider a projectile launched from ground level with an initial velocity v_0 at an angle θ above the horizontal, the motion can be described by Newton's second law. If we ask the question of which angle provides the maximum range, in the absence of air resistance, the answer is famously $45°$. However this is not the case if we include air resistance.

1.2 The Drag Force

When air resistance is included, the projectile is subject to a drag force $\mathbf{F}_r$ that always acts in the opposite direction of the velocity $\mathbf{v}$.

2 System of Differential Equations

3 Numerical Integration

4 Stopping Condition

5 How this turns into Code